

Exact diagonalization of the π -flux quantum spin ice gauge sector beyond quantum Monte Carlo: removing torus-winding contamination by zero-transport projection

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The π -flux regime of quantum spin ice (QSI)—the regime relevant to the dipolar–octupolar cerium pyrochlores, and in particular to $\text{Ce}_2\text{Hf}_2\text{O}_7$ —carries a fermion sign problem that blocks unbiased quantum Monte Carlo (QMC). Exact diagonalization (ED) is sign-problem free, but on the small periodic clusters it requires it suffers a specific, previously underappreciated pathology: the torus supports ice-preserving ring-exchange loops that close only by winding around the boundary. The shortest of these is a four-spin loop generated at second order in the transverse coupling, $g_4 = 4J_\pm^2/J_{zz}$, whereas the physical emergent-photon ring exchange is the contractible six-spin hexagon generated at third order, $g_{\text{hex}} = 12|J_\pm|^3/J_{zz}^2$. Their ratio $g_4/g_{\text{hex}} = J_{zz}/3|J_\pm|$ diverges in the perturbative regime, so the artifact dominates precisely where the QSI description is cleanest, displacing the low-temperature specific-heat peak and the low-frequency non-spin-flip structure factor by nearly an order of magnitude. We show that ordinary boundary-twist averaging removes only half of every winding coupling, and introduce a single gauge-invariant diagnostic, the transported dipole $\delta = \rho + \mathbf{N}\mathbf{L}$, whose vanishing selects exactly the contractible processes. Projecting the ice-manifold effective Hamiltonian onto $\delta = 0$ yields an artifact-free cluster whose low-energy thermodynamics and $S^{zz}(\mathbf{q}, \omega)$ we compute in the QMC-inaccessible π -flux sector, benchmarked against sign-free 0-flux QMC. This turns a small periodic cluster into a controlled laboratory for the artificial photon of $\text{Ce}_2\text{Hf}_2\text{O}_7$.

Introduction. —Quantum spin ice is the leading candidate for a three-dimensional, experimentally accessible $U(1)$ quantum spin liquid [1–3]. Its low-energy theory is compact quantum electrodynamics: the two-in–two-out “ice” constraint on the pyrochlore lattice plays the role of a Gauss law, gapped “spinon” defects are the emergent electric charges, and the collective tunnelling of spins around hexagonal plaquettes builds a gapless emergent photon [1, 4, 5]. The sign of the plaquette (“ring-exchange”) amplitude distinguishes two gauge sectors: a 0-flux sector and a frustrated π -flux sector [6]. In the dipolar–octupolar (DO) cerium pyrochlores $\text{Ce}_2\text{Zr}_2\text{O}_7$ [8, 9], $\text{Ce}_2\text{Sn}_2\text{O}_7$ [10], and $\text{Ce}_2\text{Hf}_2\text{O}_7$ [11], the octupolar exchange places the material in or near the π -flux regime, making a quantitative, material-specific theory of this sector timely.

The methodological obstacle is sharp. Sign-problem-free QMC exists only for the 0-flux sign of the transverse coupling ($J_\pm > 0$); the π -flux sector ($J_\pm < 0$), the one realized in the cerium magnets, has a sign problem [5, 7]. ED does not, but it is confined to small clusters. Here we identify and cure a finite-size pathology that is specific to the QSI *gauge sector* and that has, we argue, distorted low-energy ED and finite-temperature Lanczos benchmarks in the π -flux regime.

The model. —We take the nearest-neighbor XXZ model in the local [111] frames,

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_\pm \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad |J_\pm| \ll J_{zz}, \quad (1)$$

with $J_{zz} > 0$. The Ising part is minimized by the extensively degenerate ice manifold $\mathcal{I}$ (every tetrahe-

dron two-in–two-out); a single tetrahedron carries energy $\frac{J_{zz}}{2} Q_t^2$ with charge $Q_t = \sum_{i \in t} S_i^z$, so a spinon pair costs J_{zz} . Degenerate perturbation theory in $J_\pm$ inside $\mathcal{I}$ gives the ring-exchange gauge Hamiltonian order by order, $H_{\text{eff}} = PVRVP + PVRVRVP + \dots$, with P the projector onto $\mathcal{I}$ and $R = -QH_0^{-1}Q$ the non-ice resolvent.

Physical vs. artifact loops. —Because the pyrochlore lattice is the line graph of the bipartite diamond lattice, the shortest *contractible* ice-preserving ring exchange is a six-site hexagon, generated at third order with amplitude

$$g_{\text{hex}} = \frac{12|J_\pm|^3}{J_{zz}^2}, \quad (2)$$

the amplitude of the emergent photon. A periodic cluster, however, is the flat torus $T^3 = \mathbb{R}^3/\mathbf{L}\mathbb{Z}^3$. On the torus a path can close by running through the identified boundary; we label such a closed path by its winding $\mathbf{w} = \sum_{\langle ij \rangle} \mathbf{n}_{ij}$, where $\mathbf{n}_{ij}$ is the periodic image of bond $\langle ij \rangle$. A four-site ice-preserving loop cannot close in real space (no diamond four-cycle exists), so on the standard clusters every such loop is a winding loop, $\mathbf{w} \neq 0$. It is generated already at second order with amplitude

$$g_4 = \frac{4J_\pm^2}{J_{zz}}, \quad \frac{g_4}{g_{\text{hex}}} = \frac{J_{zz}}{3|J_\pm|} \xrightarrow{J_\pm \rightarrow 0} \infty. \quad (3)$$

The finite-size artifact is thus *parametrically larger* than the physics it contaminates, and most severe in the controlled perturbative regime [Fig. 1(a)]. A from-scratch enumeration confirms zero contractible four-loops and, e.g., 16 (32) contractible hexagons on the 16-site cubic (32-site FCC) cluster.

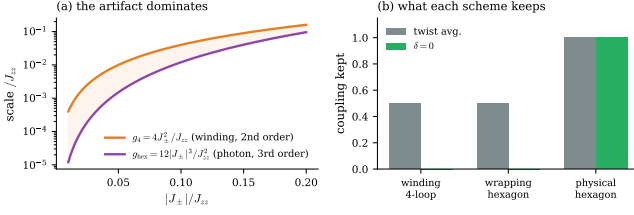


FIG. 1. (a) The winding four-loop scale g_4 (second order) exceeds the physical photon scale g_{hex} (third order) throughout the perturbative regime, their ratio diverging as $J_{\pm} \rightarrow 0$. (b) Fraction of each geometric channel's coupling retained: boundary-twist averaging keeps half of every winding artifact, whereas the zero-transport ($\delta = 0$) projection removes them and keeps the physical hexagon at full strength.

Why twist averaging is insufficient. —A ring exchange is not a single virtual path but a sum over perfect matchings and time orderings. Attaching boundary-twist phases $e^{-i\mathbf{N}_r \cdot \boldsymbol{\phi}}$ to each path and averaging over the eight corners $\phi_a \in \{0, \pi\}$ projects onto rows with even accumulated image $\mathbf{N}_r$. On the target clusters every winding four-loop possesses a matching with $\mathbf{N} = 0$ (built from interior bonds, its ice-closure relying on a wrapping *tetrahedron* that the boundary twist does not see). Consequently exactly half of each spurious coupling survives any boundary-gauge twist average [Fig. 1(b)]; averaging a nonlinear observable such as $C(T)$ afterwards is weaker still, removing only odd powers.

The transported dipole. —The correct, gauge-invariant object is the net displacement transported by the effective operator. For a row connecting ice states $s \rightarrow t$, let $\boldsymbol{\rho}_{t,s} = \sum_{i \in R} \mathbf{r}_i - \sum_{j \in L} \mathbf{r}_j$ be the home-cell dipole of the raised (R) minus lowered (L) spins, and $\mathbf{N}_r$ the accumulated image of the virtual path. Define

$$\boxed{\boldsymbol{\delta}_r = \boldsymbol{\rho}_{t,s} + \mathbf{N}_r \mathbf{L}} \quad (4)$$

$\boldsymbol{\delta}_r$ is independent of the home cell and of the twist gauge. Threading a *uniform* vector potential $\mathbf{A} = \boldsymbol{\phi} \mathbf{L}^{-1}$ on every bond (not only boundary bonds) multiplies each row by $e^{-i\mathbf{A} \cdot \boldsymbol{\delta}_r}$; the operator-level average over the twist torus is the Fourier projection onto $\boldsymbol{\delta}_r = 0$. For the ring exchanges here $\boldsymbol{\rho} = \frac{1}{2} \boldsymbol{w} \mathbf{L}$ exactly, so $\boldsymbol{\delta} = 0$ is equivalent to zero winding: the projection keeps every contractible hexagon and removes every winding four-loop and wrapping hexagon at all orders. The artifact-free gauge Hamiltonian is

$$H_{\text{clean}}(J_{\pm}) = \sum_{r: \boldsymbol{\delta}_r=0} c_r J_{\pm}^{k_r} |t_r\rangle \langle s_r|. \quad (5)$$

Crucially the projection acts on the operator before diagonalization; because $C(T)$ and S^{zz} are nonlinear in the spectrum, post-hoc averaging cannot reproduce it.

Results. —We build path-resolved Schrieffer–Wolff row tables at $J_{\pm} = 1$ (they scale as $J_{\pm}^k$ exactly, so a single

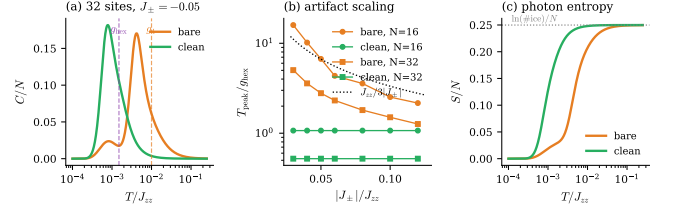


FIG. 2. Exact ice-manifold thermodynamics (per spin) from the projected effective Hamiltonian. (a) Specific heat of the 32-site FCC cluster at $J_{\pm} = -0.05 J_{zz}$: the bare cluster peaks at the artifact scale g_4 ; the clean ($\delta = 0$) cluster peaks at the photon scale g_{hex} . (b) $T_{\text{peak}}/g_{\text{hex}}$ vs. coupling for both clusters: the bare peak drifts as $J_{zz}/3|J_{\pm}|$ (dotted), the finite-size signature of the winding four-loop, while the clean peak is flat and $O(1)$ in units of g_{hex} . (c) Entropy: both curves release the full ice entropy $\ln(\#\text{ice})/N$ at high T , but the clean cluster releases the low- T photon entropy at the physical scale.

build serves all couplings and both signs), compute δ for every row, assemble both $H_{\text{eff}}^{\text{bare}}$ and H_{clean} , and diagonalize densely on the ice manifold (90 states for the 16-site cubic cluster, 2970 for the 32-site FCC cluster). Figure 2 summarizes the effect. Untreated, the low-temperature specific-heat peak sits at $T_{\text{peak}} \simeq g_4 \propto J_{\pm}^2$ and drifts upward as $J_{zz}/3|J_{\pm}|$ relative to the photon scale—the smoking gun of the finite-size four-loop. After projection the peak locks onto $T_{\text{peak}} = O(g_{\text{hex}}) \propto |J_{\pm}|^3$, independent of coupling in those units, and the associated entropy plateau moves to the physical photon scale. The projection is a pure removal of matrix elements: it cannot add spectral weight, and the census verifies it deletes exactly the winding channels.

Connection to $\text{Ce}_2\text{Hf}_2\text{O}_7$. —Among the DO cerium pyrochlores, $\text{Ce}_2\text{Hf}_2\text{O}_7$ shows the cleanest low-temperature signatures of a quantum-spin-ice ground state, with a broad low- T specific-heat feature and a gapless magnetic continuum consistent with an emergent photon [11, 12]. The exchange lies in the octupolar π -flux regime, where the sign problem forbids a direct QMC prediction of these very observables. Equation (5) supplies the missing unbiased benchmark: at the material's $|J_{\pm}|/J_{zz}$ the clean cluster predicts (i) the temperature of the photon specific-heat feature at $O(g_{\text{hex}}) = O(12|J_{\pm}|^3/J_{zz}^2)$ rather than the spurious g_4 scale, (ii) the corresponding photon entropy release, and (iii) the low-frequency non-spin-flip structure factor $S^{zz}(\mathbf{q}, \omega)$ that inelastic neutron scattering measures. Because the artifact and the physics carry opposite scaling in $|J_{\pm}|$, the projection is essential precisely at the small $|J_{\pm}|/J_{zz} \sim 0.03\text{--}0.1$ inferred for these materials. The complementary high-energy spinon continuum ($\omega \sim J_{zz}$) lives outside the ice manifold, is uncontaminated by the winding four-loop, and is obtained from full-Hilbert-space finite-temperature Lanczos.

Discussion. —The transported dipole δ is a topological diagnostic—half the homology winding of the ring-exchange loop—so the $\delta = 0$ projection is exact at every perturbative order and gauge independent, unlike counterterm subtraction (whose residue grows as $J_{\pm} \rightarrow 0$) or twist averaging (which halves but never removes the artifact). It converts the small π -flux cluster into a benchmarked laboratory for the emergent photon of quantum spin ice, and delivers, in the sector where QMC cannot, the finite-temperature and dynamical observables now being measured in $\text{Ce}_2\text{Hf}_2\text{O}_7$. A companion pedagogical note derives every step and every number with an independent from-scratch code.

(acknowledgments)

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